

Analysis 2022 Comprehensive Exam

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Problem 1. Prove that if X is a complete metric space with no isolated points, then X must be uncountable.

Proof. Suppose for contradiction that X is NOT uncountable. Since X is a complete metric space, it follows that X must be infinite. Our goal is to construct a nested sequence of closed balls

$$\overline{B}_{r_1}(y_1) \supseteq \overline{B}_{r_2} \supseteq \overline{B}_{r_3}(y_3) \supseteq \cdots$$

so that

- (1) $r_n \rightarrow 0$;
- (2) Each closed ball $\overline{B}_{r_n}(y_n)$ contains at least two points;
- (3) Each $\overline{B}_{r_n}(y_n)$ does not contain x_n for all $n \in \mathbb{N}$.

We proceed to construct the above sets via induction. Since $X \neq \emptyset$, pick a $y_1 \in X$. Since X does not contain any isolated points, we can pick another $y_2 \in X$ such that $d(y_1, y_2) < r_1$ where r_1 is small enough. If $x_1 \notin \overline{B}_{r_1}(y_1)$, then we are done. Otherwise, take $r_2 = r_1/2$ so that $x_1 \notin \overline{B}_{r_2}(y_1)$. Additionally, we see that $x_1 \notin \overline{B}_{r_2}(y_2)$. This is possible because a non-singleton ball contains points other than x_1 . Since X does not contain isolated points, there exists $y_3 \in X$ such that $d(y_2, y_3) < r_2$. If $x_2 \notin \overline{B}_{r_2}(y_2)$, then we are done. Otherwise, let $r_3 = \frac{r_2}{2} = \frac{r_1}{2^2}$ so that $x_2 \notin \overline{B}_{r_3}(y_3)$.

Inductively, suppose $\overline{B}_{r_{n-1}}(y_{n-1})$ is constructed such that $r_{n-1} = \frac{r_1}{2^{n-1}}$ and $x_{n-1} \notin \overline{B}_{r_{n-1}}(y_{n-1})$. Since $\overline{B}_{r_{n-1}}(y_{n-1})$ is not a singleton, we can choose another point $y_n \in X$ such that $d(y_n, y_{n-1}) < r_{n-1}$. Again, if $x_n \notin \overline{B}_{r_{n-1}}(y_n)$, then we are done. Otherwise, take $r_n = \frac{r_{n-1}}{2} = \frac{r_1}{2^n}$ so that $x_n \notin \overline{B}_{r_n}(y_n)$ and in turn, $x_n \notin \overline{B}_{r_{n-1}}(y_n)$. Thus, we obtain a nested sequence of closed balls where

$$\overline{B}_{r_1}(y_1) \supseteq \overline{B}_{r_2} \supseteq \overline{B}_{r_3}(y_3) \supseteq \cdots$$

so that

- (1) $r_n \rightarrow 0$;
- (2) Each closed ball $\overline{B}_{r_n}(y_n)$ contains at least two points;
- (3) Each $\overline{B}_{r_n}(y_n)$ does not contain x_n for all $n \in \mathbb{N}$.

Observe that for all $m > n$, we get that

$$d(y_n, y_m) \leq \text{diam}(\overline{B}_{r_n}(y_n)) \leq 2r_n \rightarrow 0.$$

This tells us that $\text{diam}(\overline{B}_{r_n}(y_n)) \rightarrow 0$, implying that (y_n) is a Cauchy sequence of X . Thus, $y_n \rightarrow y$ for some $y \in X$ since X is a complete metric space. By the above construction, it follows that

$$y \in \bigcap_{n=1}^{\infty} \overline{B}_{r_n}(y_n).$$

However, notice that $y \in \overline{B_{r_n}(y_n)}$ implies that $y \neq x_n$ for all $n \in \mathbb{N}$. This implies that y does not appear in any of the x_n 's inside X which is a contradiction. Thus, it follows that X must be uncountable. ■

Problem 2. Every compact metric space is separable.

Proof. Let X be a compact metric space. Our goal is to show that X is separable; that is, we need to find a countable dense subset of X . Indeed, consider the following collection of open balls of radius $1/n$

$$\mathcal{U} = \{B_{1/n}(x) : x \in X\}.$$

Notice that $\mathcal{U}$ forms an open cover of X . Since X is compact, there exists $x_1, x_2, \dots, x_{n_k} \in X$ such that

$$X = \bigcup_{k=1}^n B_{1/n}(x_{m_k}). \quad (*)$$

Now, consider the following countable union of these finite subsets of X

$$D = \bigcup_{k \in \mathbb{N}} \{x_1, x_2, \dots, x_{m_k}\}.$$

Notice that the set above is countable because each set in the union is countable. We claim that D above is a dense subset of X ; that is, $\overline{D} = X$. Let $p \in X$. To show that, $p \in \overline{D}$, it suffices to take any open ball $B_r(p)$ (with $r > 0$) such that $B_r(p) \cap D \neq \emptyset$. Indeed, since $r > 0$, we can find an $n \in \mathbb{N}$ large enough such that $\frac{1}{n} < r$. From (*), it follows that $p \in \bigcup_{i=1}^m B_{1/n}(x_i)$; that is, $p \in B_{1/n}(x_i)$ for some $1 \leq i \leq m_n$. That is,

$$d(x_i, p) < \frac{1}{n} < r.$$

But this tells us that $x_i \in B(p, r)$; that is, $B_r(p) \cap D \neq \emptyset$ (since $x_i \in D$). Thus, it follows that $p \in \overline{D}$ and we are done. ■

Problem 3. Let X be a metric space, prove that, if for all $x, y \in X$, there exists a connected set $C \subseteq X$ such that $x, y \in C$, then X is connected.

Proof. Consider the contrapositive of the above statement

"If X is NOT connected, then there exists $x, y \in X$ such that there is no connected set $C \subseteq X$ such that $x, y \in C$."

Suppose X is NOT connected. Our goal is to show that there exists $x, y \in X$ such that there is no connected set $C \subseteq X$ such that $x, y \in C$. Suppose for contradiction that there exists a connected set $C \subseteq X$ such that $x, y \in C$. Since X is NOT connected, there exists two nonempty open sets U and V in X such that $U \cap V = \emptyset$. Take $x \in U$ and $y \in V$ since U and V are nonempty sets. Since $U \cap V = \emptyset$, we have $x \notin V$ and $y \notin U$. Consider the subspace topology on C inherited from X . Note that $C \cap U$ and $C \cap V$ are open sets in this subspace. Denote these two sets as U' and V' , respectively. Notice that $U' \cup V' = C$ and that $U' \cap V' = \emptyset$ since $U \cap V = \emptyset$. However, this tells that C is a disconnected subset of X containing x and y . Thus, it follows that there exists no connected subset of X that contains x and y . ■

Problem 4. (a) Let X be a compact metric space and A be an infinite subset of X . Prove that the set of limit points of A is not empty.

Proof. Suppose for contradiction that the set of limit points A' of A is empty. Then

$$\overline{A} = A \cup A' = A \cup \emptyset = A,$$

implying that A is a closed set. Since A is a closed subset of X and X is compact, it follows that A is also a compact set. Now, since every $a \in A$, $a \in \overline{A}$, it follows that for all $r > 0$, $B_r(a) \cap A = \{a\}$. Since $\{B_r(a)\}_{a \in A}$ forms an open cover of A and A is compact, we can find $a_1, a_2, \dots, a_n \in A$ such that

$$A \subseteq \bigcup_{i=1}^n B_r(a_i).$$

Now, notice that

$$A \subseteq \left(\bigcup_{i=1}^n B_r(a_i) \right) \cap A = \bigcup_{i=1}^n [B_r(a_i) \cap A] = \bigcup_{i=1}^n \{a_i\} = \{a_i : 1 \leq i \leq n\},$$

implying that A is a finite set of X which is a contradiction. Hence, it must be that $A' \neq \emptyset$. ■

- (b) Provide a counterexample to the above statement if the compactness assumption is removed.

Solution. Suppose that $X = \mathbb{R}$ and $A = \mathbb{N} \subseteq \mathbb{R}$ which is clearly infinite but $\mathbb{N}' = \emptyset$ since there are no natural numbers contained in each open ball centered at each $n \in \mathbb{N}$. ■

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